

PREDICTION: LEGITIMATE

RISK SCORE: 54 / 100 (MEDIUM RISK)

Scam Probability: 1.04% | Model Confidence: 98.96%

Banking Action: CONTEXTUAL WARNING & 2FA CHALLENGE

Official Recommendation: Additional Verification Recommended. Some unusual characteristics were detected (elevated transfer amount or novel transfer window). Please review the recipient name, account number, and amount before submitting your biometric verification.

1. Transaction Details & Actual Parameter Values

Transaction ID:	TXN-1789631940409	Transaction Amount:	150.0
Customer ID:	C123456789	Sender Old Balance:	None
Beneficiary ID:	M987654321	Sender New Balance:	None
Payment Type:	PAYMENT	Receiver Old Balance:	None
Device Type:	None	Receiver New Balance:	None
Location Region:	None	6h Transaction Velocity:	None
New Beneficiary:	No	Prior Fraud History:	No

2. Composite Risk Score Breakdown & Latency Timing

Risk Dimension		Sub-Score	Status & Impact		
Model Fraud Probability		1.04%	Weight: 70% (Score: 0.0)		
Amount Risk		24.0 pts	Transaction size evaluation		
Velocity Risk		10.0 pts	6-hour transaction frequency		
Device Risk		10.0 pts	Device profile verification		
Location Risk		10.0 pts	Geographic regional risk		
Beneficiary Risk		65.0 pts	New beneficiary check		
Historical Behaviour		24.0 pts	Prior fraud flag check		
Balance Behaviour		0.0 pts	Account wipeout detection		
FINAL COMPOSITE RISK SCORE		54 / 100	MEDIUM RISK CATEGORY		

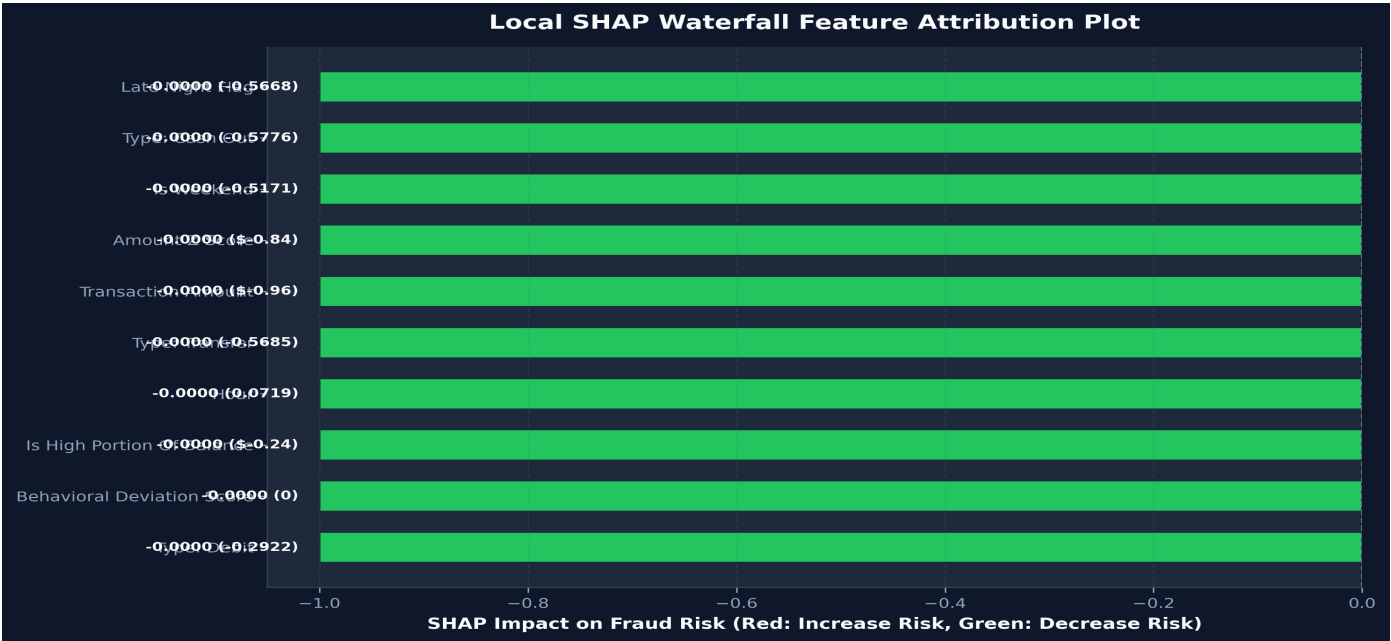
Feature Eng: 3.55 ms	Prediction: 5.19 ms	SHAP Engine: 36.09 ms	LIME Engine: 139.91 ms	Rendering: 0.00 ms	TOTAL: 1502.44 ms
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3. Banking-Grade Analyst Narrative Report

Transaction Summary:	
Behaviour Analysis:	
High Risk Indicators:	None identified.
Safe Indicators:	None identified.
Model Explanation:	
SHAP Summary:	
Final Recommendation:	

4. Top 10 SHAP Feature Contribution Attributions

Feature Name	Actual Value	SHAP Value	Contrib %	Direction	Impact Level
Transaction Amount	\$-0.96	-0.0000	3.76%	Decreases Risk	Low
Is Weekend	-0.5171	-0.0000	3.76%	Decreases Risk	Low
Late Night Flag	-0.5668	-0.0000	3.76%	Decreases Risk	Low
Type: Cash Out	-0.5776	-0.0000	3.76%	Decreases Risk	Low
Type: Transfer	-0.5685	-0.0000	3.76%	Decreases Risk	Low
Amount Z Score	\$-0.84	-0.0000	3.76%	Decreases Risk	Low
Is High Portion Of Balance	\$-0.24	-0.0000	3.76%	Decreases Risk	Low
Hour	0.0719	-0.0000	3.76%	Decreases Risk	Low
Behavioral Deviation Score	0	-0.0000	3.76%	Decreases Risk	Low
Amount To Oldbalance Ratio	\$-0.31	-0.0000	3.76%	Decreases Risk	Low



5. Local LIME Decision Rules & Explanations

Decision Rule Condition	Rule Weight	Effect Direction	Domain Explanation
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Account Balance Wipeout = Yes (100% Drained)	+0.2800	Increases Risk	Sender account balance drained to zero or near-zero post-transfer.
amount <= -0.96	+0.0000	Decreases Risk	Transaction amount is within low-risk customer spending bounds.
log_amount <= -1.84	+0.0000	Decreases Risk	Transaction amount is within low-risk customer spending bounds.
oldbalanceOrg <= -1.43	+0.0000	Decreases Risk	Sender account maintains sufficient baseline capital.
log_oldbalanceOrg <= -1.56	+0.0000	Decreases Risk	Sender account maintains sufficient baseline capital.
amount_to_oldbalance_ratio <= -0.31	+0.0000	Decreases Risk	Transaction amount is within low-risk customer spending bounds.
is_high_portion_of_balance <= -0.24	+0.0000	Decreases Risk	Feature condition 'is_high_portion_of_balance <= -0.24' supported legitimate transaction verification.
hour <= 0.07	+0.0000	Decreases Risk	Feature condition 'hour <= 0.07' supported legitimate transaction verification.
day_of_week <= -1.43	+0.0000	Decreases Risk	Feature condition 'day_of_week <= -1.43' supported legitimate transaction verification.
is_weekend <= -0.52	+0.0000	Decreases Risk	Feature condition 'is_weekend <= -0.52' supported legitimate transaction verification.

